

MCQs and Descriptive Questions Generation for PDFs and MCQs generation for videos

Introduction:

This project is for generating MCQs and descriptive questions out of a PDF. The user selects the number of MCQs and descriptive questions for each chapter, and based on that the number of MCQs and descriptive questions are generated for each chapter. The user also gets options to whether to choose all the chapters of the PDF or for only specified chapter.

Similarly, the other project is to generate MCQs out of a video. The user uploads a video, and a summary is generated out of the given video, and then the user selects the number of MCQs to be generated. Those number of MCQs are generated for that particular video.

Content Types:

Contents	Page
Title	1
Introduction:	1
Technology:	2
Key features:	2
User Manual	2-10

Technology:

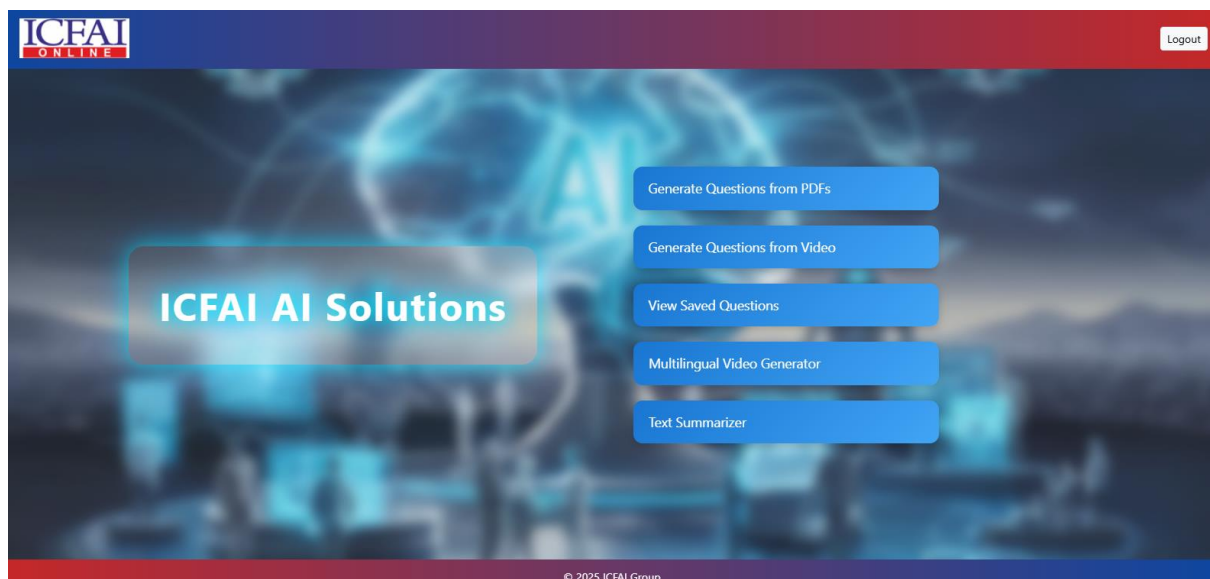
Layer	Technology
Backend	Python, Llama 3
Frontend	HTML, CSS, and JavaScript
Database	MySQL
Authentication	Session based login
Server	Localhost (development environment)

Key Features:

- To generate MCQs and descriptive questions for a PDF and MCQs for a video.
- To generate descriptive answers as well as answer for a n MCQ.
- Generated questions are level based from 1-5
- There are options to download the questions as a document, excel and a csv file.
- The user can generate a question paper out of the questions generated by giving the number of questions to be generated for each level.

Main Page:

This is the main page. Generate Questions from PDFs icon is for generating MCQs and descriptive questions from PDFs. Generate questions from video is for generating MCQS from the video. View saved questions is for viewing the database entries and generating question paper out of those entries for each level. Multilingual video generator and text summarizer are different projects which lead to different pages.



Generate Questions From a PDF:

The user uploads a PDF and gets to choose whether to generate the number of MCQs or Descriptive questions or both. He also gets to choose whether to generate questions using AI model or using content from the text book. After choosing all the required options the user clicks on generate MCQS button to generate the questions and answers too.

PDFs uploaded
0

Pages (current PDF)
0

MCQs generated
0

Descriptive Qs
0


Drag and drop file here
Limit 200MB per file • PDF

Choose File: Statistics for Business and Economics, Tenth Edition (David R. Anderson, Dennis J. Sweeney etc.) (z-lib.org).pdf

Found 145 TOC entries across 22 chapters.
Detected 145 TOC entries across 22 chapters — Pages detected: 1054

Scope

☐ All topics ☒ Selected chapters only

MCQ source
From textbook (use chapter content)

Question type
Both

MCQs per topic
1

Descriptive per topic
1

Generate MCQs

Chapters (pick for "Selected chapters only")

☐ Chapter 1 • 6 topics	☐ Chapter 2 • 6 topics	☐ Chapter 3 • 6 topics	☐ Chapter 4 • 4 topics
☐ Chapter 5 • 8 topics	☐ Chapter 6 • 6 topics	☐ Chapter 7 • 11 topics	☐ Chapter 8 • 6 topics
☐ Chapter 9 • 9 topics	☒ Chapter 10 • 2 topics	☐ Chapter 11 • 4 topics	☐ Chapter 12 • 5 topics
☐ Chapter 13 • 7 topics	☐ Chapter 14 • 12 topics	☐ Chapter 15 • 11 topics	☐ Chapter 16 • 6 topics

Generated Output From a PDF:

This is how the output comes in the page based on the selected questions for each topic of that particular chapter. The user can download the generated questions and answers as a PDF or an Excel or a CSV.

Inferences About Two Populations Using Minitab (MCQs: 1 • Descr: 1)

Multiple Choice Questions

1. Q1. A clinical trial is conducted to compare the average blood pressure of patients taking a new drug versus those taking a placebo. The sample mean blood pressure for the new drug group is 120 mmHg with a standard deviation of 10 mmHg, while the sample mean blood pressure for the placebo group is 130 mmHg with a standard deviation of 12 mmHg. Assuming that the true population means are normally distributed and equal to μ_1 and μ_2 , respectively, which of the following statements is a correct conclusion based on this data? — Level 4

- A. The difference between the two population means is likely to be due to chance.
- B. There is strong evidence to suggest that the new drug has caused a significant reduction in blood pressure compared to the placebo.
- C. Since both sample standard deviations are greater than 5, we can conclude that the true population standard deviations (σ_1 and σ_2) are likely to be less than 10 mmHg.
- D. The confidence interval for $\mu_1 - \mu_2$ contains only positive values. Answer: B Difficulty: 4

✓ Answer: B

Descriptive / Short-answer Questions

1. Suppose a pharmaceutical company wants to compare the mean time it takes to heal from an infection using two different antibiotics, A and B. The company collects data on the healing times for each antibiotic and finds that the sample mean healing time for antibiotic A is 5.2 days with a standard deviation of 0.8 days, while the sample mean healing time for antibiotic B is 4.5 days with a standard deviation of 0.9 days. Assuming equal variances, what can be inferred about the difference in mean healing times between the two antibiotics? — Level 4

✓ Answer: Based on the given data, it can be concluded that there is a statistically significant difference in mean healing times between the two antibiotics, with antibiotic B resulting in shorter mean healing times.

Inferences About Two Populations Using Excel (MCQs: 1 • Descr: 1)

Multiple Choice Questions

1. Q1. An interval estimate for the difference between two population means is constructed using which of the following formulas? — Level 3

- A. $(\bar{x}_1 - \bar{x}_2) \pm t * (s_1^2 / n_1 + s_2^2 / n_2)$
- B. $(\bar{x}_1 - \bar{x}_2) \pm t * \sqrt{(s_1^2 / n_1 + s_2^2 / n_2)}$
- C. $(\bar{x}_1 - \bar{x}_2) \pm t * (s_1^2 / n_1 - s_2^2 / n_2)$
- D. $(\bar{x}_1 - \bar{x}_2) \pm t * (n_1 + n_2)$ Answer: B Difficulty: 3

Generated for 2 topics — MCQs: 2 — Descriptive: 2

Download DOCX

Download Excel

Download CSV

 Export to Database

Generate Questions From a Video:

The user uploads a video and selects the transcribe button to get the text from the video and generate a summary from the transcription. Once the transcript is generated a summary is generated from it and the output looks like how it is given in the below image.

Videos uploaded

0

MCQs generated

0

Drag and drop file here

Limit 200MB per file • MP4, MOV, MKV, AVI, MPEG4

Choose File

Synchronicity in Innovation (1).mp4

Transcribe

Generate MCQs

Selected file: Synchronicity in Innovation (1).mp4 (187310 KB)

Transcript will appear here (or paste one)...

Question type

MCQs

Number of MCQs

10

Transcribe

Generate MCQs

Transcription complete.

Hello everyone. Today I'm going to talk about synchronicity in innovation with so many new things coming up in today's world, new technologies, for example, generative AI, it's very important to understand what is synchronicity and why it matters. So let's understand what synchronicity is and how it is important in innovation. So what I'm going to cover in today's presentation are the following topics. I will cover why timing matters, the idea of multiples, strength of weak ties, absorptive capacity, cadences and gates, ecosystem orchestration, open innovation, real options, logic and measurement. So I think these are important. So let's discuss them one by one. Now why timing and alignment matters? Let's look at this first. So synchronicity is basically purposeful alignment in time across ideas, teams and ecosystems that accelerates discovery and commercialization. So there has to be an alignment across time over ideas, teams and ecosystems. And it should be used for commercializing, otherwise it's a waste of time. So what I'm saying here is leaders should create conditions which support this. How can leaders create conditions where valuable coincidences are more likely? So you can also think of synchronicity as alignment of clocks. In innovation, this means when an

Summary

synchronicity is purposeful alignment in time across ideas, teams and ecosystems that accelerates discovery and commercialization. In innovation, this means when an idea, a technology and the market all come together at the right moment. Even brilliant ideas can fail if the timing is wrong. The idea of multiples, strength of weak ties, absorptive capacity, cadences and gates, ecosystem orchestration, open innovation, real options, logic and measurement are among the topics I'll cover.

A firm's ability to recognize the value of the new external information assimilated into existing knowledge and apply it commercially is called absorptive capacity. Nacodac invented the digital camera in 1975 but didn't have the capacity to pivot from film to digital. IBM successfully absorbed knowledge from universities and startups during the AI revolution, turning Watson into healthcare and finance product line. Google is a very good example of absorptivecapacity. Then comes process design, cadences and gates. To be expressed align with fast moving customer requirements.

Open innovation as a synchronizer, getting ideas from the public, that is important. ecosystem orchestration, which is important, aligning compliments. Tesla supercharger rollouts synchronized EV adoption. Microsoft synchronized Azure cloud expansion with partner ecosystem, consulting firms, ISVs so that when enterprise moved, compliments were ready. It's a deliberate way to create synchronicity rather than relying on chance. Over synchronization can also backfire, and there are problems with over synchronizaton. So if you over synchronize a business in one area, that might lead to lagging in another area.

Leaders must balance synchrony with exploratory Slack like Google's 20% time, Intel, TikTok. Cadence worked for years, but became brittle when Moore's law slowed. So similarly, companies can think of practical 90 day plan. For example, what Spotify does, it aligns product release calendars with major music industry events like Grammy Awards. In 90 days, leaders can identify one anchor event in their industry and align internal cadences to it. Simultaneously, set up a weak type brokerage group for apps cross-functional innovation council that meets monthly to surface opportunities for synchrony.

Generated Output From a Video:

This is the output for the MCQs generated for the video after the summary is generated from the video.

Question type

MCQs ▾

Number of MCQs

1|

Download DOCX

Download Excel

Download CSV

MCQs

1. What is the concept of synchronicity in innovation?

A. Purposeful alignment of ideas across teams and ecosystems

B. Random combination of technologies and markets

C. Slow adaptation to changing customer requirements

D. Overemphasis on internal research and development

✓ Answer: A

View Saved Questions:

When the user clicks this icon, it shows the database entries in it and provides the option for the user to generate a question paper by giving the number of questions for each level.

Saved Questions in Database Home

Search by keyword... All Types Refresh

Found 222 questions

ID	Topic	Type	Question	Options/Answer	Difficulty	Created
219	Inferences About Two Populations Using Minitab	MCQ	Q1. A clinical trial is conducted to compare the average blood pressure of patients taking a new drug versus those taking a placebo. The sample mean blood pressure for the new drug group is 120 mmHg with a standard deviation of 10 mmHg, while the sample mean blood pressure for the placebo group is 130 mmHg with a standard deviation of 12 mmHg. Assuming that the true population means are normally distributed and equal to μ_1 and μ_2 , respectively, which of the following statements is a correct conclusion based on this data?	A: A. The difference between the two population means is likely to be due to chance. B: B. There is strong evidence to suggest that the new drug has caused a significant reduction in blood pressure compared to the placebo. C: C. Since both sample standard deviations are greater than 5, we can conclude that the true population standard deviations (σ_1 and σ_2) are likely to be less than 10 mmHg. D: D. The confidence interval for $\mu_1 - \mu_2$ contains only positive values. Answer: B Difficulty: 4 Answer: B	4	10/16/2025
220	Inferences About Two Populations Using Minitab	Descriptive	Suppose a pharmaceutical company wants to compare the mean time it takes to heal from an infection using two different antibiotics, A and B. The company collects	Answer: Based on the given data, it can be concluded that there is a statistically significant difference in mean healing times between the two antibiotics, with antibiotic B resulting in shorter mean healing times.	4	10/16/2025

Generate Question Paper

Level 1: Level 2: Level 3: Level 4: Level 5:

Generated Question Paper (3 questions)

Question	Answer
1 Describe the steps in Excel to construct a 95% confidence interval for the difference between two population means ($\mu_1 - \mu_2$) when the population standard deviations (σ_1 and σ_2) are known, assuming independent random samples from each	